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Unit – 1 : Session – 3 : SLO – 1 | Nested Structures

■ I. Fill in the Blanks

1. inside — a nested structure means a structure inside another structure.
2. struct — the struct keyword defines the nested structure.
3. member (inner member name) — outer.inner.member.
4. multiple (two or more) — one dot per nesting level.
5. nested structure.
6. variable — memory is allocated when the outer structure variable is declared.
7. inner (nested) — B is the inner/nested structure.
8. related (logically grouped) — nested structures group related sets of information.
9. FALSE — age is not a member of address; the correct path depends on the design, e.g. student.age or student.dob.date.
10. the dot operator applied repeatedly, outer to inner (outer.inner.member).

■ II. Code Snippets

1. Student/address: `struct Address { char city[30]; int pincode; }; struct Student { char name[50]; struct Address addr; }; — access: s.addr.city.`
2. Employee/salary: `struct Salary { float basic, bonus; }; struct Emp { char name[50]; struct Salary sal; }; — e.sal.basic.`
3. Book/author: `struct Author { char name[50]; char country[20]; }; struct Book { char title[50]; struct Author auth; }; — b.auth.name.`
4. Car/engine: `struct Engine { int capacity; char type[20]; }; struct Car { char brand[20], model[20]; struct Engine eng; }; — c.eng.capacity.`
5. Hospital/contact: `struct Contact { char phone[15]; char email[40]; }; struct Patient { char name[50]; int age; struct Contact con; }; — p.con.phone.`

■ III. Practice the Code

- `struct Date { int day, month, year; };`
- `struct Employee { char name[50]; int id; struct Date joining; };`
- `struct Employee e = {"Ravi", 101, {12, 8, 2023}};`
- `printf("Joined on %d-%d-%d", e.joining.day, e.joining.month, e.joining.year);`